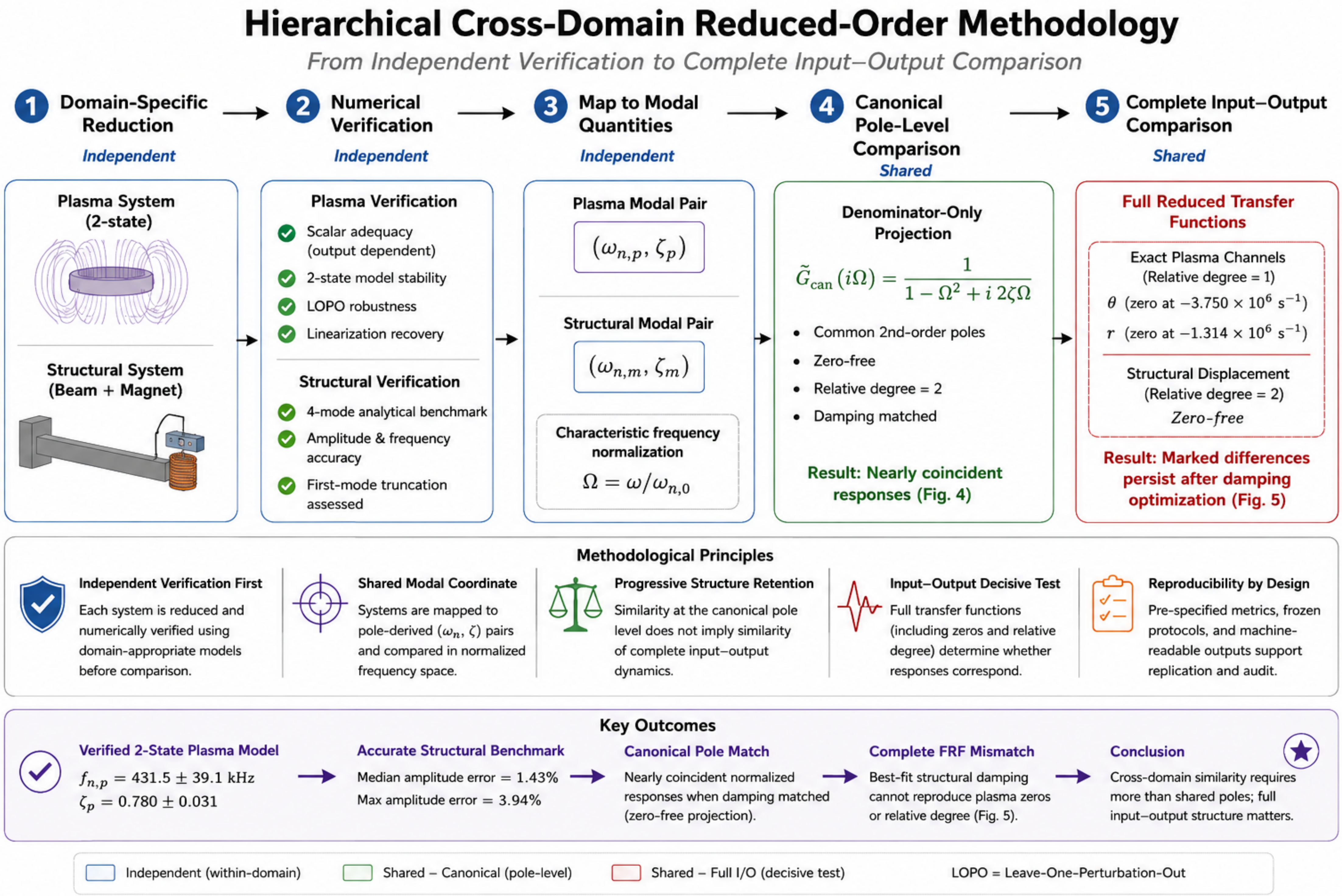
Hierarchical Cross-Domain Reduced-Order Methodology
From Independent Verification to Complete Input–Output Comparison
1 Domain-Specific Reduction
Independent
2 Numerical Verification
Independent
3 Map to Modal Quantities
Independent
4 Canonical Pole-Level Comparison
Shared
5 Complete Input–Output Comparison
Shared
Plasma System (2-state)
Structural System (Beam + Magnet)
Plasma Verification
Scalar adequacy (output dependent)
2-state model stability
LOPO robustness
Linearization recovery
Structural Verification
4-mode analytical benchmark
Amplitude & frequency accuracy
First-mode truncation assessed
Plasma Modal Pair
$(\omega_{n,p}, \zeta_p)$
Structural Modal Pair
$(\omega_{n,m}, \zeta_m)$
Characteristic frequency normalization
$\Omega = \omega/\omega_{n,0}$
Denominator-Only Projection
$\tilde{G}_{\mathrm{can}}(i\Omega) = \dfrac{1}{1 - \Omega^2 + i\,2\zeta\Omega}$
Common 2nd-order poles
Zero-free
Relative degree = 2
Damping matched
Result: Nearly coincident responses (Fig. 4)
Full Reduced Transfer Functions
Exact Plasma Channels
(Relative degree = 1)
θ (zero at $-3.750 \times 10^6\ \mathrm{s}^{-1}$)
r (zero at $-1.314 \times 10^6\ \mathrm{s}^{-1}$)
Structural Displacement
(Relative degree = 2)
Zero-free
Result: Marked differences persist after damping optimization (Fig. 5)
Methodological Principles
Independent Verification First
Each system is reduced and numerically verified using domain-appropriate models before comparison.
Shared Modal Coordinate
Systems are mapped to pole-derived (ω_n, ζ) pairs and compared in normalized frequency space.
Progressive Structure Retention
Similarity at the canonical pole level does not imply similarity of complete input–output dynamics.
Input–Output Decisive Test
Full transfer functions (including zeros and relative degree) determine whether responses correspond.
Reproducibility by Design
Pre-specified metrics, frozen protocols, and machine-readable outputs support replication and audit.
Key Outcomes
Verified 2-State Plasma Model
$f_{n,p} = 431.5 \pm 39.1$ kHz
$\zeta_p = 0.780 \pm 0.031$
Accurate Structural Benchmark
Median amplitude error = 1.43%
Max amplitude error = 3.94%
Canonical Pole Match
Nearly coincident normalized responses when damping matched (zero-free projection).
Complete FRF Mismatch
Best-fit structural damping cannot reproduce plasma zeros or relative degree (Fig. 5).
Conclusion
Cross-domain similarity requires more than shared poles; full input–output structure matters.
Independent (within-domain)
Shared – Canonical (pole-level)
Shared – Full I/O (decisive test)
LOPO = Leave-One-Perturbation-Out